
JavaScript Notes – Deep Version (Part 1)

[Without DOM & Arrays]

1. JavaScript – Ek Nazar Mein (Theory)

- **JavaScript** ek **client-side scripting language** hai.
 - Browser ke andar chalta hai.
 - Web page ko **interactive** banane ke liye use hota hai.
 - **JavaScript + HTML + CSS = Full Web Page**
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2. Variables – var, let, const

Variables ka kaam hota hai **data store** karna.

◆ **var:**

- Pehle ka use hota tha
- Function scoped
- Re-declare bhi ho jata hai

◆ **let:**

- Modern JavaScript (ES6)
- Block scoped
- Re-declare nahi hota

◆ **const:**

- Fixed value
- Change nahi kar sakte

```
var a = 10;
let b = 20;
const c = 30;

console.log(a, b, c);
```

📌 Output:

```
10 20 30
```

✓ 3. Data Types – Detailed

JavaScript me 2 type ke data types hote hain:

◆ Primitive Types:

- String
- Number
- Boolean
- Null
- Undefined
- Symbol
- BigInt

◆ Non-Primitive (Reference):

- Object
 - Array
 - Function
-

📊 Null vs Undefined

Topic	Null	Undefined
Value	Khud se set karte ho	Default hota hai
Type	Object	Undefined
Example	<code>let a = null;</code>	<code>let b;</code>

```
let a = null;
let b;

console.log(a); // null
console.log(b); // undefined
```

📌 Output:

```
null
undefined
```

✓ 4. Operators

◆ Arithmetic

```
let x = 10, y = 5;
console.log(x + y); // 15
console.log(x - y); // 5
console.log(x * y); // 50
```

◆ Comparison

```
console.log(5 == "5"); // true
console.log(5 === "5"); // false
```

◆ Logical

```
console.log(true && false); // false
console.log(true || false); // true
```

✓ 5. Conditional Statements

◆ if-else

```
let marks = 80;

if (marks >= 90) {
  console.log("Grade A");
} else if (marks >= 70) {
  console.log("Grade B");
} else {
  console.log("Fail");
}
```

⬆️ Output:

Grade B

◆ Switch Case

Agar multiple choices ho, to switch use karte hain.

```
let day = 3;

switch(day) {
  case 1:
    console.log("Monday");
    break;
  case 2:
    console.log("Tuesday");
    break;
  case 3:
    console.log("Wednesday");
    break;
  default:
    console.log("Invalid Day");
}
```

⬆️ Output:

Wednesday

🔔 **Note:** Har case ke baad break lagana zaroori hai warna sab case challenge.

✓ 6. Loops – Repeat ka system

◆ for loop

```
for (let i = 1; i <= 3; i++) {
  console.log("Loop:", i);
}
```

⬆️ Output:

Loop: 1
Loop: 2
Loop: 3

◆ while loop

```
let i = 1;
while(i <= 3) {
    console.log("While Loop:", i);
    i++;
}
```

📌 Output:

```
While Loop: 1
While Loop: 2
While Loop: 3
```

✓ 7. Functions – Logic ek jagah

◆ Function banane ka tareeka

```
function greet(name) {
    console.log("Hello " + name);
}

greet("Amit");
```

📌 Output:

```
Hello Amit
```

✓ 8. Objects – Real Life Data

```
let car = {
    brand: "Toyota",
    model: "Fortuner",
    year: 2023
};

console.log(car.brand);
```

📌 Output:

```
Toyota
```

✓ 9. typeof – Data Type Check Karne Wala

```
let x = 42;
let y = "Hello";
let z = null;

console.log(typeof x); // number
console.log(typeof y); // string
console.log(typeof z); // object !
```

📌 Output:

```
number
string
object ← (null ka type object hota hai JavaScript ka purana bug)
```

✓ 10. Hoisting – JavaScript ka magic

JavaScript code ko **top se read** karta hai. Variables aur functions ko upar utha leta hai (hoist karta hai).

```
console.log(a); // undefined
var a = 10;
```

📌 Output:

```
undefined
```

Let & Const hoist hote hain but use se pehle error dete hain.

🏠 Summary:

Topic	Done? ✓
Variables	✓
Data Types	✓
if/else, switch	✓
Loops	✓
Functions	✓
Objects	✓
typeof, null vs undefined	✓
